IOT-BASED REAL-TIME MOTION ACCIDENT DETECTION AND GPS TRACKING SYSTEM WITH GUI-INTEGRATED TRANSMITTER AND COMMUNICATION

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CHAPTER I: INTRODUCTION

1.1 Background of the Study

The World Health Organization (2023) reports that road traffic injuries remain a leading cause of death globally, claiming approximately 1.19 million lives annually. A significant factor in these fatalities is the "Golden Hour," the immediate period following a traumatic injury where prompt medical intervention significantly increases survival rates. However, in many instances, victims are incapacitated or located in remote areas where manual emergency reporting is impossible. This gap in safety infrastructure has led to the development of automated accident detection systems.

Existing solutions primarily utilize Global System for Mobile Communications (GSM) modules to send SMS alerts. While effective in urban settings, GSM-based systems face significant limitations, including reliance on cellular network availability, high power consumption, and recurring operational costs. Furthermore, these systems often lack a user-side interface, leaving the operator unaware of the device's operational status or sensor health. The integration of Internet of Things (IoT) technologies and Micro-Electro-Mechanical Systems (MEMS) offers a more resilient alternative. By combining high-precision accelerometers with localized Radio Frequency (RF) communication, it is possible to create a dedicated safety link that functions independently of cellular infrastructure. The addition of an embedded Graphical User Interface (GUI) enhances the system by providing real-time feedback, allowing for manual arming/disarming and system health monitoring, thereby reducing false positives and increasing user confidence in the safety device.

1.2 Conceptual Framework

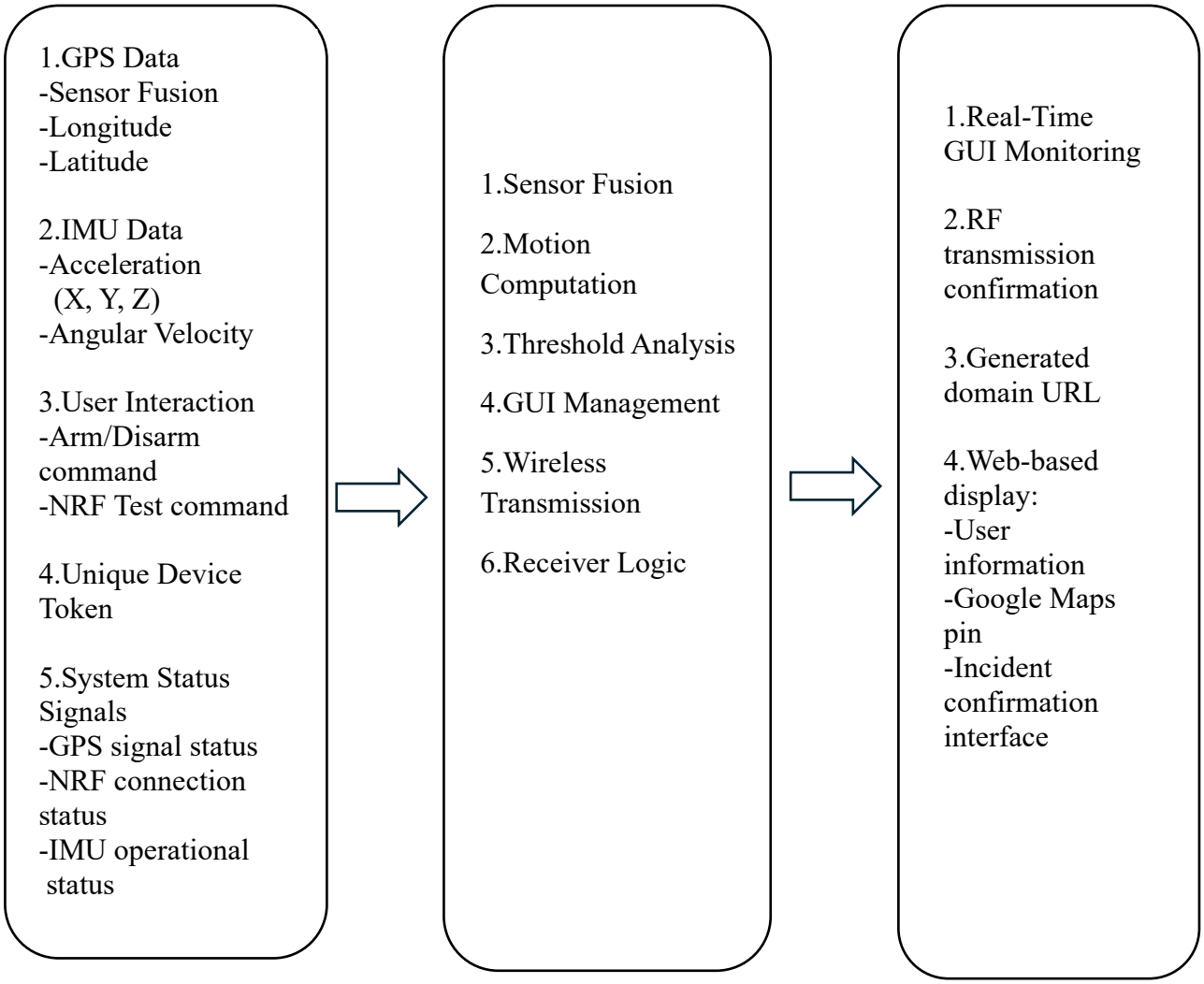

Figure 1. Conceptual Framework

The conceptual framework of this study is structured around the Input–Process–Output (IPO) model, detailing the flow from raw data acquisition to final emergency verification.

The input stage encompasses the raw data streams and user-defined parameters required for system operation. This includes geospatial data (longitude and latitude) from the NEO-7M GPS module and tri-axial motion data (acceleration and angular velocity) from the MPU6050. User-side inputs include interactive commands such as Arm/Disarm and NRF connectivity

tests. Furthermore, the system inputs include a hard-coded Unique Device Token and diagnostic signals representing the operational status of the GPS lock, RF link, and IMU calibration.

The data acquisition and processing stage transforms raw sensor inputs into meaningful emergency information through a layered computational approach. The microcontroller first establishes synchronized communication with the MPU6050 via the I2C protocol for high-speed motion sampling and with the NEO-7M GPS module through UART for parsing NMEA coordinate data. Using the acquired tri-axial acceleration values, the system computes the resultant vector magnitude to determine total G-force impact levels, expressed as:

$$A=\sqrt{x^2 + y^2 + z^2}.$$

Simultaneously, orientation angles are calculated to monitor spatial tilt and instability, given by

$$\text{Pitch} = \tan^{-1}\left(\frac{x}{\sqrt{y^2+z^2}}\right)$$

and

$$\text{Roll}= \tan^{-1}\left(\frac{y}{\sqrt{x^2+z^2}}\right).$$

These computed values undergo threshold analysis, where abnormal exponential spikes exceeding calibrated safety limits are classified as potential accidents. Once an impact event is confirmed, the processed data—comprising the unique device token, GPS coordinates, and motion status flag—is encapsulated into an SPI-driven RF packet and transmitted via the NRF24L01 module. On the receiver side, the Arduino Pro Micro decodes the packet and executes Human Interface Device (HID) emulation to automatically generate the corresponding verification URL, completing the hardware-to-web emergency reporting cycle.

The output stage provides the final deliverables of the system. Locally, the user sees real-time monitoring and status confirmations on the transmitter GUI. Remotely, the system generates a dynamic domain URL: https://jiecep-csusc/{unique_token}/{longitude}/{latitude}/. This URL launches a web-based interface displaying the registered user information and a Google Maps pin, facilitating immediate incident confirmation and dispatch.

1.3 Objectives of the Study

The overall objective of this study is to design and develop a real-time motion accident detection and monitoring system with an integrated GUI-based transmitter interface and RF-based remote verification.

Specifically, the study aims to:

1. To implement real-time GPS monitoring using the NEO-7M module.
2. To compute pitch and roll from IMU data for orientation visualization.
3. To detect abnormal motion events using threshold-based vector analysis.
4. To design a transmitter-side GUI for local data visualization.
5. To integrate NRF testing and arming/disarming mechanisms within the GUI.
6. To implement a token-based identification system for secure device mapping.
7. To develop a receiver-side HID emulation system for automated URL generation.

1.5 Statement of the Problem

The general problem of this research is the lack of a low-cost, real-time accident monitoring system that provides both local user feedback and remote emergency verification without cellular dependency.

Specifically, the study seeks to answer:

1. How can the system minimize false-positive triggers during routine movement?
2. How can real-time orientation (pitch and roll) be visualized to enhance user-side monitoring?
3. How reliable is the NRF24L01 communication in maintaining a stable link for emergency packets?
4. How can a unique identification system be implemented to prevent data overlap between multiple devices?
5. How can the receiver automate web-based reporting without requiring specialized software on the host computer?

1.6 Scope and Limitations of the Study

The study focuses on localized monitoring with a range of approximately 100 to 800 meters depending on the NRF24L01 antenna type. The GUI is constrained by the resolution of the OLED/TFT display used and the processing overhead of the Arduino. GPS accuracy is subject to clear line-of-sight to satellites, and the system is designed for ground-based vehicular or personal fall incidents.

1.7 Significance of the Study

This study is significant as it offers a faster and more reliable way to detect accidents and send real-time location alerts. Its features are expected to benefit several groups involved in safety and emergency response. The following beneficiaries include:

EMERGENCY RESPONDERS: Real-time alerts enable faster and more efficient rescues.

COMMUTERS AND ROAD USERS: Added safety through automatic detection for cyclists and motor riders.

DRIVERS AND MOTORISTS: Enhanced protection in high-traffic or isolated areas.

ISOLATED INDIVIDUALS: Automated assistance for elderly or solo travelers.

LOCAL EMERGENCY AGENCIES: Improved dispatch strategies and resource allocation.

THE ECE COMMUNITY: Demonstrates the viability of license-free RF bands for life-safety applications.

1.8 Definition of Terms

The following terms are defined for a better understanding of this study:

PITCH- The rotation of the device around its lateral axis.

ROLL- The rotation of the device around its longitudinal axis.

GUI- The visual interface on the transmitter for user interaction.

ARMING MODE- The state where the system actively monitors for impact triggers.

TOKEN-BASED AUTHENTICATION- Use of a unique identifier to link hardware to a specific user database.

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